AMCS Textiles Ltd.

Plot # 41 & 52,AEPZ,Adamjee Nagar,Siddhirganj,Narayanganj-1400
(23.679648N, 90.521818E)
18th May 2014



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Observations

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Load bearing capacity of steel connections

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Steel connections main structure:

There are gaps (10-15mm wide) between the head plates of connections in various locations and sometimes deformations of flanges at connections without stiffeners.

Connection capacities should be checked by a structural engineer.

Load bearing capacity of steel connections

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Steel connections stairs:

The connections of the external staircases to the main structure appear to be non-engineered. The connections of the internal stairs appear to be weak.

Structural engineer should check connection capacities.

Load bearing capacity of steel connections

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Missing or loose bracing

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Roof bracing connection to beam.

Roof bracing connection to beam is at an angle and not in line with bracing. Potential for the connection to break.

Bracing on 3rd floor is slack in places.



Bracing on 3rd floor.

No apparent diagonal bracing in 3rd floor at perimeter walls (at the location where it occurs at the other floors levels).

Bracing

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Apparently Non-engineered Steel Mezzanine Floor

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Non-Engineered Mezzanine under heavy loading.

Mezzanine floor on Ground floor appears to be non-engineered and is currently being used for storage.
Slender channel columns used under heavy loading.
Connections between steel members are welded.





Connections between steel members

Apparently Non-Engineered Steel Mezzanine Floor

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High storage loads

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Heavy rolls of material stacked approx. 3m high. (Ground Floor level).
It was unclear if the ground floor slab is suspended onto grade beams/pile caps or is supported by the ground.





High storage loads on steel mezzanine floor

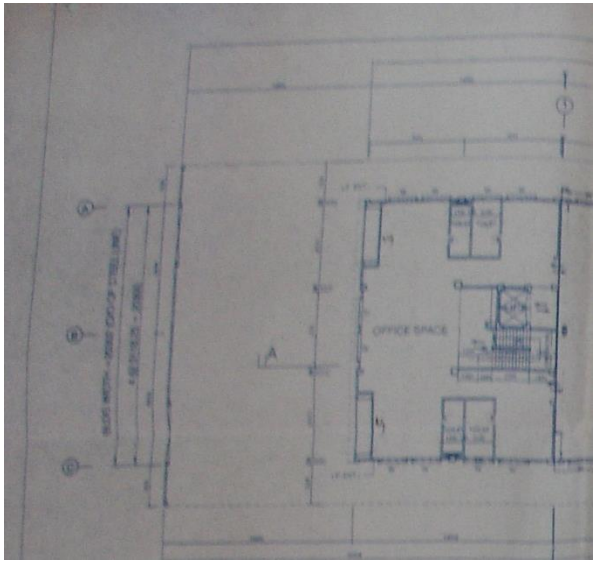
High Storage Loads

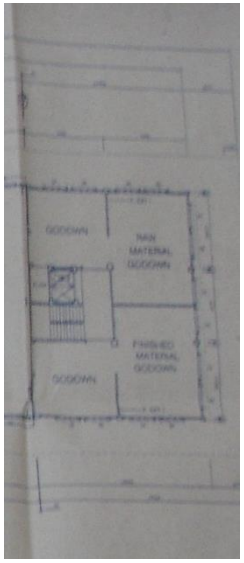
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Inconsistencies with permit drawings

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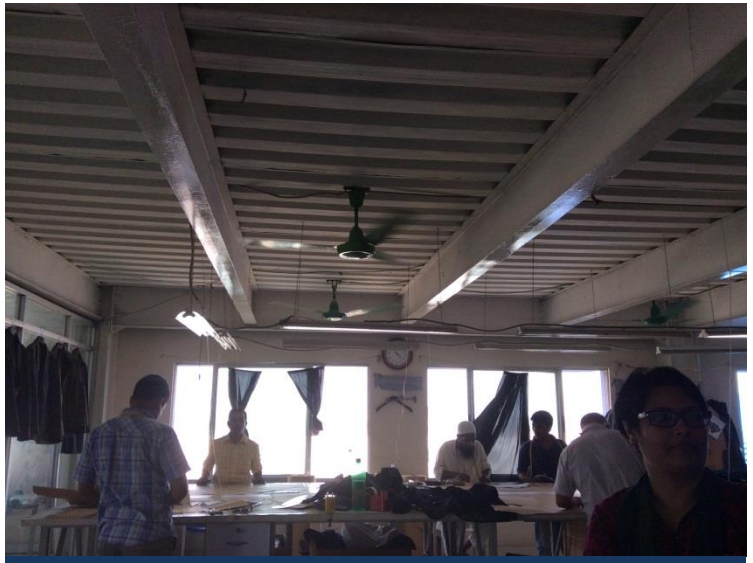




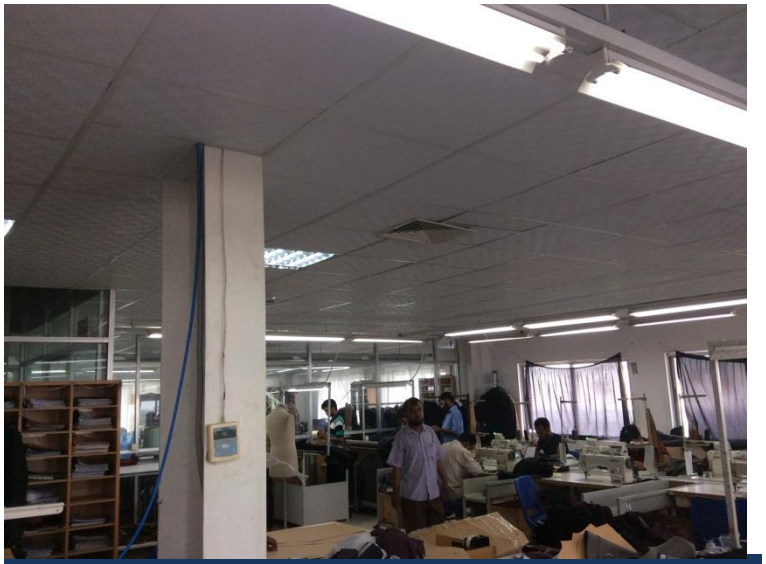
2nd and 3rd floor office block permit drawings showing concrete structure

Permit drawings showed permit for concrete office blocks at either end of the building.

2nd and 3rd floor sections are steel structures.

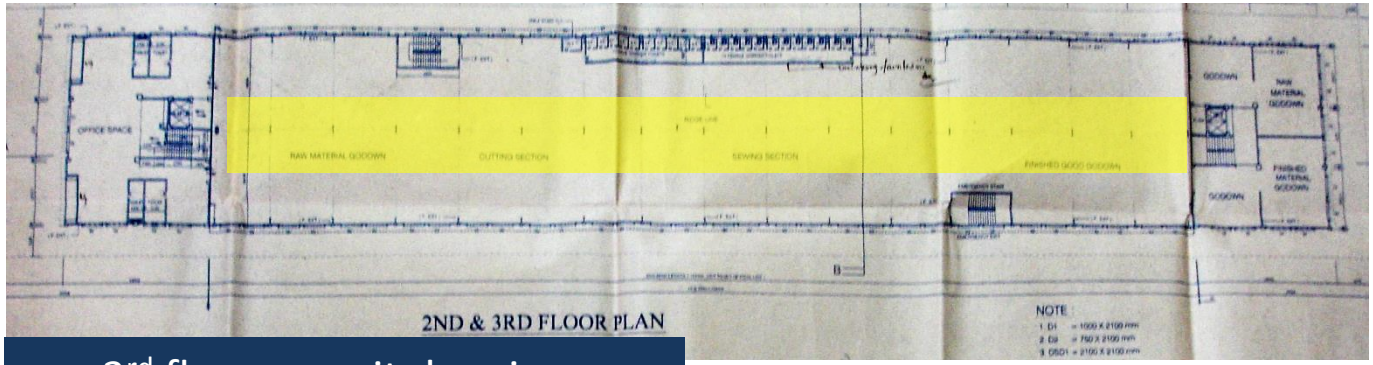


3rd floor steel structure

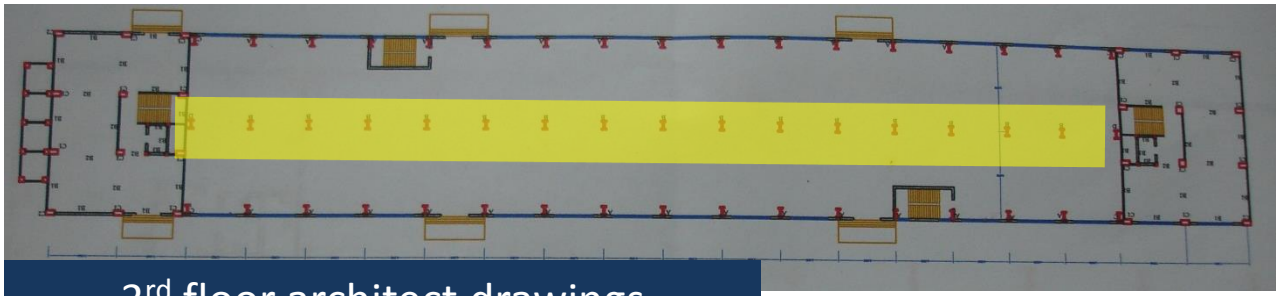


2nd floor steel structure

Inconsistencies with permit drawings



3rd floor permit drawings



3rd floor architect drawings

Permit & architect's drawings show a column line down the centre of the 3rd floor steel structure. However, the middle columns have not been constructed.



Inconsistencies with permit drawings

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Priority Actions

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Problems Observed

ITEM 1: Load bearing capacity of steel connections

ITEM 2: Missing or not tightened bracing

ITEM 3: Non-engineered steel mezzanine floor

ITEM 4: High storage loads

ITEM 5: Inconsistencies with permit drawings

Item No.	Observation	Recommended Action Plan	Recommended Timeline
1	Load bearing capacity of steel connections	Structural Engineer to check the connections capacities of all steel connections.	**6-weeks**
2	Load bearing capacity of steel connections	Repairs and alterations to be made to the steel connections as required and as advised by the Structural Engineer.	**6-weeks**
3	Missing or not tightened bracing	Structural Engineer to assess the slack roof bracing and the missing diagonal bracing in the 3rd floor.	**6-weeks**
4	Missing or not tightened bracing	Repairs to be carried out as required.	**6-weeks**
5	Non-engineered steel mezzanine floor.	Structural Engineer to provide detailed calculations for the mezzanine floor. These should confirm structure's ability to support storage loading and its extends.	**6-weeks**

Item No.	Observation	Recommended Action Plan	Recommended Timeline
6	High storage loads	Structural engineer to confirm loadbearing system at Ground Floor slab level	**6-weeks**
7	High storage loads	Produce and actively manage a loading plan for all floor plates including ground floor and mezzanine floors within the factory giving consideration to floor capacity and column capacity	**6-weeks**
8	High storage loads	Continue to implement load plan.	**6-months**
9	Inconsistencies with permit drawings	The Building Engineer to check that the as-built structure (no central column row in 3rd floor and steel structure for office blocks in 2nd and 3rd floor) is in alignment with the structural design.	**6-weeks**
10	Inconsistencies with permit drawings	Carry out any structural alterations as advised by the Structural engineer.	**6-weeks**